Notes: Davis (RFS, 2025)
The Elasticity of Quantitative Investment

Contents

1 Big picture

- **Question.** What is the price elasticity of demand for statistical arbitrageurs that invest using modern cross-sectional asset pricing models?
- **Core contrast.** Classical no-arbitrage and APT-style models often rely on statistical arbitrageurs who trade aggressively against price movements, producing highly elastic asset demand. Davis shows that many realistic quantitative portfolio choice models imply strikingly inelastic demand.
- **Definition of statistical arbitrageur.** The paper studies investors who form portfolios from predictors, factor models, or machine-learning models to maximize expected return per unit of volatility, sometimes hedging systematic risk.
- **Main empirical object.** The object is not only the portfolio weight from a model, but how that weight changes when a stock price changes.
- **Data.** The paper combines 13F institutional holdings, CRSP returns, 62 stock predictors from the Freyberger-Neuhierl-Weber data, and the Kenneth French risk-free rate. The main out-of-sample elasticity tests run from February 1990 to January 2020.
- **Classical benchmark.** A simple mean-variance calibration gives a value-weighted elasticity around 900. This means a 1% price drop would imply a very large increase in shares demanded.
- **Main result.** Thirteen statistical arbitrage models from the literature generate far lower elasticities, typically around 0 to 10 after value-weighting and winsorization.
- **Why inelastic?** Quantitative strategies diversify across noisy predictors rather than aggressively buying every stock that falls in price. Trading against price changes is difficult because price signals are noisy and because portfolio weights depend on cross-sectional predictors.
- **Pure-alpha robustness.** Hedging out factor exposures does not turn the portfolios into elastic demand. The pure-alpha strategies remain inelastic.
- **Demand-system exercise.** A quantitative equilibrium model that adds these high-Sharpe-ratio statistical arbitrageurs to the market shows that aggregate demand remains inelastic.
- **Alpha implication.** The inelasticity of the statistical arbitrageurs helps explain why factor-model alphas can persist: the arbitrageurs are not elastic enough to arbitrage them away completely.
- **Economic takeaway.** The paper challenges the common intuition that better asset-pricing models automatically create highly elastic arbitrage capital. Better models can still imply conservative, diversified, and inelastic demand curves.

2 Data and measurement

2.1 Data sources

- **Institutional holdings.** Quarterly long-only institutional holdings come from SEC 13F filings from 1984 to 2020. Following Koijen and Yogo, the paper adds a residual "household" investor to hold shares not accounted for in 13F filings.
- **Returns and shares.** CRSP provides monthly returns, changes in shares outstanding, dividends, and stock prices.
- **Stock predictors.** The paper uses 62 monthly predictors from Freyberger, Neuhierl, and Weber, extended through 2020 by Baba-Yara, Boons, and Tamoni.
- **Risk-free rate.** The risk-free rate is the standard one-month Treasury bill rate from Kenneth French's website.
- **Main out-of-sample period.** The main empirical results use 1,253,175 stock-month observations from February 1990 through January 2020.

Interpretation: The data are designed to measure both portfolio choice and the price sensitivity of that portfolio choice. Holdings data tell who owns what; predictor and return data tell how quantitative models form demand.

2.2 Notation

Let $P_{i,t}$ denote market equity of stock i at time t, $p_{i,t}$ the share price, and $S_{i,t}$ shares outstanding:

$$P_{i,t} = S_{i,t}p_{i,t}.$$

Let $r_{i,t}$ be the excess return of stock i and let N_t be the number of stocks in the period.

Interpretation: Elasticity is defined in terms of shares demanded, but the portfolio choice models usually output portfolio weights. The distinction between shares, prices, and weights is central.

2.3 Raw and normalized predictors

Let $x_{i,k,t}$ be raw predictor k for stock i. The paper transforms raw predictors into normalized predictors $\dot{z}_{i,k,t}$. A discontinuous percentile transformation would be

$$\dot{z}_{i,k,t} = \frac{\text{rank}(x_{k,t})_i - 1}{N - 1} - 0.5.$$

Because elasticity requires derivatives with respect to prices, Davis uses a continuous version:

$$\dot{z}_{i,k,t} = K(\text{ArcSinh}(x_{k,t}))_i - 0.5,$$

where $K(\cdot)$ is the cumulative distribution function of a standard kernel density function.

Interpretation: The continuous transformation preserves the economic idea of percentile-sorted predictors while allowing the derivative $\partial \dot{z}_{i,k,t}/\partial \log(p_{i,t})$ to be calculated.

2.4 Price-based and non-price-based predictors

- Non-price-based predictors are treated as exogenous to contemporaneous price changes. Examples include historical beta, book values, asset values, investment, and profitability.
- Price-based predictors change mechanically when prices change. Examples include book-to-market, earnings-to-market, debt-to-price, Tobin's q, payout-to-price, and short-term reversal.
- The paper identifies 15 of the 62 predictors as endogenous to prices.

Interpretation: Only price-based predictors contribute directly to demand elasticity. If a model invests in predictors that do not move with price, its weights do not react to price and its demand is less elastic.

2.5 Predictor-weighted portfolios

The zero-cost return associated with predictor k is

$$\widehat{F}_{k,t+1} = \sum_{i=1}^{N} \dot{z}_{i,k,t} r_{i,t+1}.$$

The paper rescales predictors so leverage in predictor-weighted portfolios does not mechanically vary with the number of available stocks.

Interpretation: Predictor-weighted portfolios are the building blocks of the statistical arbitrage models. They turn stock characteristics into tradable factor returns.

3 Models and empirical specifications

3.1 Elasticity of shares demanded

Let a_t be assets under management and $w_{i,t}$ the portfolio weight on stock i. Shares demanded are

$$s_{i,t} = \frac{a_t w_{i,t}}{p_{i,t}}.$$

The demand elasticity is the negative log change in shares demanded from a log price change:

$$\eta_{i,t} = -\frac{\partial \log(s_{i,t})}{\partial \log(p_{i,t})} = 1 - \frac{\partial \log(w_{i,t})}{\partial \log(p_{i,t})} - \frac{\partial \log(a_t)}{\partial \log(p_{i,t})}. \tag{1}$$

Interpretation: The first term is the mechanical fact that if the investor keeps dollar weight fixed, shares fall when price rises. The second term captures how the strategy changes portfolio weights when price changes. The third term is the wealth effect.

3.2 Elasticity when wealth effects are negligible

The paper shows wealth effects are small for diversified models, so the main elasticity is approximated as

$$\eta_{i,t} = 1 - \frac{1}{w_{i,t}} \frac{\partial w_{i,t}}{\partial \log(p_{i,t})}. \tag{2}$$

Interpretation: If a falling price leads the model to increase its weight, the derivative is negative and elasticity rises. If the model lowers weight after the price falls, elasticity can be near zero or even negative.

3.3 Classical mean-variance benchmark

A classical statistical arbitrageur with CARA utility and multivariate normal returns has demand

$$w_t = \frac{1}{\gamma a_t} \Sigma_t^{-1} \mu_t, \tag{3}$$

where γ is risk aversion, a_t is AUM, Σ_t is the covariance matrix, and μ_t is expected excess returns.

Using the Stevens decomposition, demand for asset i can be written as

$$w_{i,t} = \frac{1}{\gamma a_t} \frac{\mu_{i,t} - \beta'_{i,t}\mu_{-i,t}}{\sigma^2_{i,t,\epsilon}}. \tag{4}$$

This implies the calibrated elasticity

$$\eta_{i,t} = 1 + \frac{1}{w_{i,t}} \frac{1}{\gamma a_t \sigma^2_{i,t,\epsilon}} \left(-\frac{\partial \mu_{i,t}}{\partial \log(p_{i,t})} \right). \tag{5}$$

Plugging in the paper's estimates gives approximately

$$\eta_{i,t} \approx 1 + \frac{0.27}{w_{i,t}}.$$

Interpretation: This calibration produces very high elasticities for most stocks because market weights are small. A typical stock with weight around 1/3481 has elasticity near 900.

3.4 Statistical arbitrage models

Stack normalized predictors into the $N \times K$ matrix Z_t. Many statistical arbitrage models can be written as

$$w_t = f(Z_t)b,$$

where $f(Z_t)$ maps predictors into candidate portfolio weights and b collapses the candidate portfolios into the final strategy.

Examples include:

- linear predictor-weighted portfolios, $w_t = Z_t b$;
- Fama-French and Hou-Xue-Zhang style factor portfolios;
- Kozak-Nagel-Santosh, Kelly-Pruitt-Su, and Gu-Kelly-Xiu portfolios;
- random forests and neural networks.

Interpretation: These models try to maximize Sharpe ratios or price the cross-section. The central question is how strongly their final weights change when the underlying stock price changes.

3.5 Elasticity of statistical arbitrage models

For assets with positive weights, Davis writes

$$\eta_{i,t} = 1 - \frac{1}{w_{i,t}} \sum_{k=1}^{K} \nabla_{z_{i,k,t}} (f(Z_t)_i b) \frac{\partial z_{i,k,t}}{\partial \log(p_{i,t})}. \tag{6}$$

Interpretation: Elasticity is the product of two forces: how much predictors move with price, and how strongly the model invests according to those predictors.

3.6 Elasticity for long and short positions

For short positions, the paper uses absolute position size:

$$\eta^{\pm}_{i,t} = 1 - \frac{1}{|w_{i,t}|} \frac{\partial w_{i,t}}{\partial \log(p_{i,t})}. \tag{7}$$

Interpretation: This keeps the interpretation of the slope of the demand curve while avoiding undefined log percentage changes for negative positions.

3.7 Pure-alpha hedging

To isolate pure-alpha investment, the paper hedges a strategy using another right-hand-side factor portfolio:

$$w_{a,t} = w_t - W_t \beta. \tag{8}$$

The elasticity becomes

$$\frac{\partial w_{a,i,t}}{\partial \log(p_{i,t})} = \frac{\partial w_{i,t}}{\partial \log(p_{i,t})} - \sum_j \frac{\partial W_{i,j,t}}{\partial \log(p_{i,t})} \beta_j. \tag{9}$$

Interpretation: If low elasticity is only due to systematic risk exposure, hedging should make pure-alpha strategies more elastic. The paper finds it does not.

3.8 Demand system and equilibrium

The aggregate demand system is written as

$$w_{i,t} = \beta_{i,0,t} + \beta_{i,1,t}\frac{P_{i,t}}{A_t} + \beta_{i,2,t}\log(P_{i,t}). \tag{10}$$

Equilibrium requires aggregate shares demanded to equal shares outstanding:

$$\frac{A_t}{p_{i,t}}\left(\beta_{i,0,t} + \beta_{i,1,t}\frac{P_{i,t}}{A_t} + \beta_{i,2,t}\log(P_{i,t})\right) = S_{i,t}. \tag{11}$$

The resulting aggregate elasticity is

$$\eta_{i,t} = 1 - \beta_{i,1,t} - \frac{\beta_{i,2,t}}{P_{i,t}/A_t}. \tag{12}$$

Interpretation: The demand-system exercise embeds statistical arbitrageurs into an equilibrium demand framework. It asks whether adding them changes aggregate market elasticity.

3.9 Aggregate elasticity with many investors

Let $D_{i,t} = \sum_j |D^j_{i,t}|$ be total absolute dollars invested in asset i across investors, and let $P_{i,t}$ be market equity. Under a zero-position regularity condition,

$$\eta^{agg}_{i,t} = 1 + \frac{D_{i,t}}{P_{i,t}}\left(\bar{\eta}_{i,t} - 1\right), \qquad \bar{\eta}_{i,t} = \frac{1}{D_{i,t}}\sum_{j\in J_i} D^j_{i,t}\eta^{j,\pm}_{i,t}. \tag{13}$$

Interpretation: Aggregate market elasticity is close to a position-size-weighted average of investor elasticities. Inelastic statistical arbitrageurs therefore do not automatically make aggregate demand elastic.

4 Empirical design and interpretation

4.1 What identifies the elasticity estimates

- The paper computes model-implied portfolio weights from fully specified portfolio choice rules.
- It differentiates those weights with respect to log prices through the price-sensitive predictors.
- The out-of-sample model parameters are fit ex ante using expanding windows.
- Standard errors are double-clustered by month and stock in the main elasticity tables.

Interpretation: This is not a reduced-form flow-price regression. It is a model-implied elasticity calculation: if a quantitative strategy follows this rule, its demand curve has this slope.

4.2 Why the benchmark matters

- The classical benchmark produces elasticity around 900 because expected returns are assumed to rise strongly enough after price drops and because weights are small.
- The benchmark demonstrates how elastic statistical arbitrage demand would look in a frictionless mean-variance world.
- The empirical models then show how modern quantitative strategies depart from that ideal.

Interpretation: The gap between 900 and roughly 0 to 10 is the core empirical finding.

4.3 Why price-based predictors are crucial

- If a predictor does not change with price, it cannot make the model change weight when price changes.
- Price-based predictors such as valuation ratios do change with price, but their effect on demand depends on whether the model uses them aggressively.
- Diversified models often spread exposure across many predictors and stocks, limiting sensitivity to any one price change.

Interpretation: Quantitative models can have high Sharpe ratios but still be inelastic if their weights do not respond aggressively to idiosyncratic price movements.

4.4 Why hedging matters

- The paper tests whether inelasticity comes from systematic factor exposure rather than pure-alpha positioning.
- It forms hedged portfolios by using the other model portfolios as right-hand-side factors.
- The hedged pure-alpha portfolios continue to have low elasticity.

Interpretation: The result is not just that factor models are systematic risk portfolios. Even after hedging, the pure-alpha component is not highly elastic.

4.5 Counterfactual equilibrium exercise

- The paper estimates a demand system that accommodates high-Sharpe-ratio statistical arbitrageurs.
- It gives statistical arbitrageurs counterfactual market shares up to 25%.
- Prices are endogenized, while shares outstanding, dividends, and non-price predictors are treated as exogenous.
- Statistical arbitrageur parameters are recursively refit using endogenous return histories.
- The key result is that aggregate elasticity changes little, typically by between about -0.2 and +0.2.

Interpretation: Even if these strategies manage meaningful capital, their own inelastic demand prevents them from making the market highly elastic.

4.6 What the paper cannot fully prove

- The paper studies model-implied demand from academic portfolio rules, which may differ from proprietary real-world quantitative trading systems.
- The counterfactual demand system requires assumptions about how non-HSA investors behave when HSAs enter the market.
- Derivatives for discontinuous models such as random forests require numerical approximations.
- The paper focuses on statistical arbitrage elasticity, not every possible form of liquidity provision or high-frequency market making.

Interpretation: The evidence is strongest as a critique of the assumption that modern cross-sectional quantitative investment automatically generates classical no-arbitrage elasticity.

5 Tables and figures

5.1 Figure 1 and Table 1: demand mapping and price-sensitive predictors

Read:

- Figure 1 maps prices into raw predictors, raw predictors into normalized predictors, and normalized predictors into portfolio weights.
- Some predictors mechanically depend on prices; others do not.
- Table 1 reports the 15 price-sensitive predictors and the median derivative of each predictor with respect to log price.
- Valuation ratios with price in the denominator generally have negative derivatives; Tobin's q has a positive derivative.

Economic message: Elasticity can only arise through the path from prices to predictors to weights. Table 1 quantifies the first step in that path.

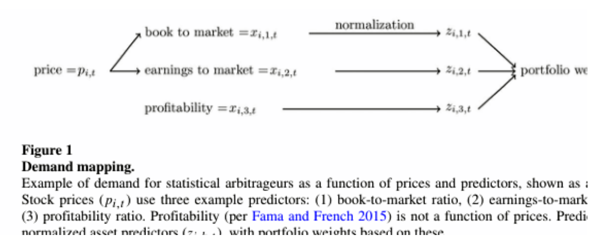

Figure 1: Demand mapping.

Table 1
Predictor price sensitivity.

	$\frac{\partial z_{i,k,t}}{\partial \log(p_{i,t})}$		$\frac{\partial z_{i,k,t}}{\partial \log(p_{i,t})}$		$\frac{\partial z_{i,k,t}}{\partial \log(p_{i,t})}$
Covariates	Gradient	Covariates	Gradient	Covariates	Gradient
a2me	−0.273	e2p	−0.248	o2p	−0.125
	(−0.422, −0.082)		(−0.534, 0.056)		(−0.275, −0.000)
beme	−0.374	ldp	$< 10^{-10}$	q	0.321
	(−0.551, −0.083)		(−0.296, −0.000)		(0.076, 0.833)
beme_adj	−0.343	lme	0.147	rel_to_high_price	1.28
	(−0.691, −0.064)		(0.057, 0.184)		(0.287, 3.070)
cum_return_1_0	2.93	lme_adj	0.037	roc	0.161
	(0.504, 5.114)		(0.003, 0.192)		(0.019, 0.662)
debt2p	−0.174	nop	−0.061	s2p	−0.274
	(−0.258, −0.000)		(−0.248, 0.072)		(−0.389, −0.062)
Obs.	1,869,963		1,869,963		1,869,963
Months	703		703		703

Median (across assets and months) of $\partial z_{i,k,t}/\partial \log(p_{i,t})$, showing how predictors change with a 1% increase in asset prices. Also, 10th and 90th percentiles are in parentheses (not confidence intervals). Variable descriptions from Freyberger, Neuhierl, and Weber (2020) include: a2me (asset to market equity), beme (book-to-market), beme_adj (book-to-market minus industry mean), cum_return_1_0 (short-term reversals), debt2p (debt to price), e2p (earnings to price), ldp (dividend to price), lme (size), lme_adj (size minus industry size), nop (net payout to price), o2p (payout to price), q (Tobin's q), rel_to_high_price (52-week high ratio), roc (rents over cash), and s2p (sales to price).

Table 1: Predictor price sensitivity.

5.2 Figure 2 and Table 2: classical demand calibration

Read:

- Figure 2 shows that calibrated classical elasticities decline with market portfolio weights.
- The smallest stocks have extremely high elasticities; the largest stocks still have elasticities around 10.
- Table 2 reports equal-weighted elasticities above 44,000 even after winsorization.
- The value-weighted elasticity is about 929 under all winsorization schemes.

Economic message: Classical statistical arbitrage demand is far more elastic than demand estimated in the empirical asset demand literature.

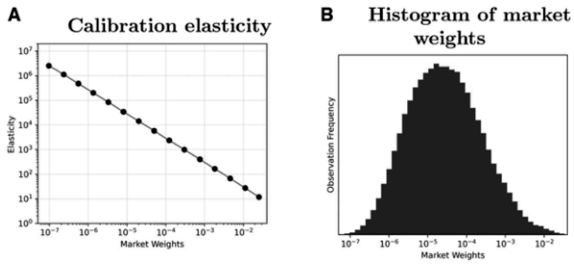

Figure 2
Elasticity and market weights.
Panel A: Binned scatterplot of calibrated elasticities versus market portfolio weights (both axes in log-scale).
Panel B: Histogram of market portfolio weights (log-scale x-axis). Based on 1,253,175 stock-month observations

Figure 2: Elasticity and market weights.

Table 2
Demand elasticity calibration.

	Average elasticity		
	(1)	(2)	(3)
Winsorization:	None	(1^{st},99^{th})	(5^{th},95^{th})
Equal weighted	61,157.8***	56,004.5***	44,679.4***
	(1,966.1)	(1,740.7)	(1,287.8)
Value weighted	929.4***	929.2***	928.9***
	(19.1)	(19.1)	(19.1)
Largest stock	10.8***		
	(0.3)		

Calibrated elasticities under different weighting and winsorization schemes. Averages are reweighted, winsorized, then averaged cross-sectionally and over time. Rows: equal-weighted, value-weighted (market cap shares), and largest stock in each cross-section. Columns: no winsorization, 1st–99th percentile winsorization, and 5th–95th percentile winsorization. Based on 1,253,175 stock-month observations from February 1990 to January 2020. Standard errors (in parentheses) are double-clustered by month and stock.*p<0.1; **p<0.05; ***p<0.01.

Table 2: Demand elasticity calibration.

5.3 Tables 3 and 4: model returns and model-implied elasticities

Read:

- Table 3 shows that the 13 statistical arbitrageur portfolios have large out-of-sample CAPM alphas and high Sharpe ratios.
- Table 4 is the paper's central elasticity table.
- The value-weighted, 5th-95th percentile winsorized elasticities are much lower than 929.
- Examples: BPZ_F is 0.142, BSV is 1.833, FF3 is 0.858, HXZ is roughly zero, and GKX is 9.433.

Economic message: These models can perform well as return predictors while still producing inelastic demand. High alpha does not imply high price elasticity.

Table 3
Statistical arbitrageur returns.

	BPZ_F	BPZ_L	BSV	CRW	DGU	FF3	FF6	GKX	HXZ	KNS	KPS	NN	RF
	(1)	(2)	(3)	(4)	(5)	(6)	(7)	(8)	(9)	(10)	(11)	(12)	(13)
α	21.526***	33.489***	32.940***	9.615***	14.094***	9.044***	24.950***	50.879***	11.338***	36.955***	18.289***	20.638***	9.267***
	(2.520)	(2.592)	(2.628)	(2.598)	(2.576)	(2.379)	(2.601)	(2.620)	(2.325)	(2.625)	(2.593)	(2.373)	(2.281)
β	0.285***	-0.165***	-0.007	-0.153***	0.199***	0.425***	0.143***	-0.077	0.466***	-0.049	0.163***	0.430***	0.497***
	(0.051)	(0.052)	(0.053)	(0.052)	(0.052)	(0.048)	(0.052)	(0.053)	(0.047)	(0.053)	(0.052)	(0.048)	(0.046)
Sharpe	1.681	2.267	2.318	0.591	1.107	0.881	1.841	3.543	1.066	2.577	1.383	1.701	0.937
Obs.	360	360	360	360	360	360	360	360	360	360	360	360	360

Monthly CAPM annualized alpha, betas, and Sharpe ratios for the 13 statistical arbitrageurs from February 1990 to January 2020 (out-of-sample period). Standard errors are in parentheses. See Table A.1 for model initialisms.*p<0.1; **p<0.05; ***p<0.01.

Table 3: Statistical arbitrageur returns.

Table 4
Statistical arbitrageur price elasticity.

Elasticity Weighting	Wins.	BPZ$_F$ (1)	BPZ$_L$ (2)	BSV (3)	CRW (4)	DGU (5)	FF3 (6)	FF6 (7)	GKX (8)	HXZ (9)	KNS (10)	KPS (11)	NN (12)	RF (13)
		Average elasticity												
Portfolio	None	0.285***	6.428***	2.275***	2.034***	3.209***	1.911***	1.553***	7.394***	0.843***	3.630***	2.951***	2.496***	2.185***
		(0.094)	(0.190)	(0.093)	(0.057)	(0.437)	(0.022)	(0.017)	(0.552)	(0.012)	(0.049)	(0.504)	(0.595)	(0.037)
	(5^{th},95^{th})	-1.233***	6.252***	2.354***	2.019***	2.753***	1.969***	1.561***	7.923***	0.866***	3.600***	2.545***	1.893***	1.864***
		(0.082)	(0.100)	(0.035)	(0.047)	(0.034)	(0.021)	(0.016)	(0.096)	(0.011)	(0.045)	(0.290)	(0.069)	(0.032)
Value	None	12.781***	36.835***	7.105***	17.635**	25.920**	3.003***	0.290	76.991***	-1.208***	13.109***	27.588***	13.516***	2.101***
		(1.291)	(3.281)	(0.886)	(8.063)	(11.499)	(0.704)	(0.594)	(25.222)	(0.114)	(1.301)	(4.917)	(2.954)	(0.142)
	(5^{th},95^{th})	0.142**	7.168***	1.833***	1.964***	2.203***	0.858***	0.706***	9.433***	-0.004	3.253***	3.510***	2.163***	0.955***
		(0.059)	(0.266)	(0.071)	(0.085)	(0.087)	(0.026)	(0.021)	(0.311)	(0.007)	(0.117)	(0.476)	(0.152)	(0.033)

Average elasticities of the 13 models. Averages are reweighted, winsorized, then averaged cross-sectionally and over time. Weighting schemes: portfolio weights and value weights (market cap). Results are either not winsorized or winsorized at the 5th–95th percentiles. Based on 1,253,175 stock-month observations from February 1990 to January 2020. Standard errors (in parentheses) are double-clustered by month and stock. See Table A.1 for model initialisms.*p<0.1; **p<0.05; ***p<0.01.

Table 4: Statistical arbitrageur price elasticity.

5.4 Table 5 and Figure 3: including long and short positions

Read:

- Table 5 compares the positive-weight sample with the full long-short sample using the absolute-position elasticity formula.
- Elasticities are similar when short positions are included.
- Figure 3 compares the statistical arbitrageur elasticities against the classical calibrated elasticity across market weights.
- In most size bins, the calibrated elasticity line is far above the statistical arbitrageur lines.

Economic message: The main result is not an artifact of ignoring shorts. The models remain much more inelastic than the classical benchmark.

Table 5
Statistical arbitrageur price elasticity by sample.

Elasticity Sample	BPZ$_F$ (1)	BPZ$_L$ (2)	BSV (3)	CRW (4)	DGU (5)	FF3 (6)	FF6 (7)	GKX (8)	HXZ (9)	KNS (10)	KPS (11)	NN (12)	RF (13)
	Average elasticity												
Positive	0.142**	7.168***	1.833***	1.964***	2.203***	0.858***	0.706***	9.433***	-0.004	3.253***	3.510***	2.163***	0.955***
	(0.059)	(0.266)	(0.071)	(0.085)	(0.087)	(0.026)	(0.021)	(0.311)	(0.007)	(0.117)	(0.476)	(0.152)	(0.033)
All	0.447***	7.946***	2.074***	2.027***	3.005***	1.217***	0.923***	10.060***	0.168***	3.812***	3.535***	1.737***	0.959***
	(0.031)	(0.263)	(0.071)	(0.088)	(0.094)	(0.028)	(0.022)	(0.308)	(0.005)	(0.117)	(0.497)	(0.113)	(0.032)

Average price elasticity of the 13 models for two samples: (1) positive weights and (2) all assets, using Equation (10). Results are value-weighted and winsorized (5th–95th percentiles). Sample: 1,253,175 stock-month observations, February 1990 to January 2020. Standard errors (in parentheses) are double-clustered by month and stock. See Table A.1 for model initialisms.*p<0.1; **p<0.05; ***p<0.01.

Table 5: Elasticity by sample.

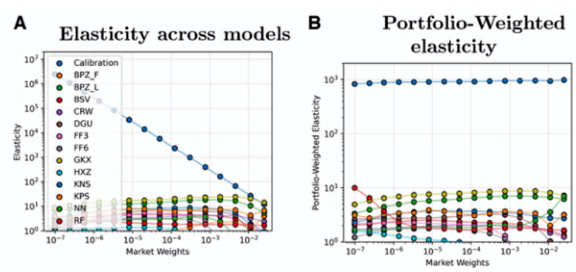

Figure 3
Statistical arbitrageur versus calibrated elasticity.
Binned scatterplots of elasticities versus market portfolio weights (log-scale axes), winsorized in each bin (5th–95th percentiles). Panel A: Elasticity; Panel B: Portfolio-weighted elasticity. Sample: 1,253,175 stock-month

Figure 3: Statistical arbitrageur versus calibrated elasticity.

5.5 Tables 6 and 7: pure-alpha portfolios and hedged elasticity

Read:

- Table 6 reports annualized alphas after hedging each model portfolio by the other model portfolios.

- Many entries remain economically large, indicating that much of the model return survives as pure alpha relative to other models.
- Table 7 reports elasticities of those hedged portfolios.
- Hedged elasticities are still generally small in magnitude and do not resemble the classical elasticity benchmark.

Economic message: Inelasticity is not only a feature of systematic factor exposure. Even pure-alpha versions of the strategies are not highly elastic.

Table 6
Cross alphas.

Rows correspond to left-hand-side portfolios, columns correspond to right-hand-side portfolios

	BPZ_F (1)	BPZ_L (2)	BSV (3)	CRW (4)	DGU (5)	FF3 (6)	FF6 (7)	GKX (8)	HXZ (9)	KNS (10)	KPS (11)	NN (12)	RF (13)
BPZ_F		13.156***	6.744**	24.970***	16.994***	19.321***	20.292***	7.736**	20.047***	5.046*	22.391***	13.564***	24.085***
		(2.924)	(2.667)	(2.606)	(2.451)	(2.495)	(2.910)	(3.516)	(2.627)	(2.774)	(2.785)	(2.615)	(2.686)
BPZ_L	24.237***		9.208***	31.644***	29.479***	33.574***	24.290***	-0.667	33.630***	1.337	31.901***	24.771***	33.487***
	(2.719)		(2.236)	(2.626)	(2.683)	(2.658)	(2.801)	(2.811)	(2.700)	(1.739)	(2.792)	(2.751)	(2.673)
BSV	20.494***	10.446***		33.675***	28.025***	29.144***	27.763***	3.829	32.569***	-0.193	34.418***	20.249***	34.138***
	(2.463)	(2.221)		(2.619)	(2.589)	(2.553)	(2.880)	(3.029)	(2.713)	(1.377)	(2.784)	(2.462)	(2.675)
CRW	11.606***	6.423**	11.486***		13.776***	13.004***	9.117***	1.912	13.383***	9.857***	8.822***	12.256***	9.195***
	(2.856)	(3.095)	(3.108)		(2.557)	(2.486)	(2.936)	(3.681)	(2.560)	(3.231)	(2.792)	(2.852)	(2.681)
DGU	5.322**	10.229***	5.535*	18.585***		6.982***	2.199	13.078***	2.795**	5.275*	11.418***	4.136	11.304***
	(2.595)	(3.055)	(2.968)	(2.470)		(1.916)	(2.514)	(3.707)	(1.413)	(3.099)	(2.725)	(2.536)	(2.535)
FF3	3.884	16.151***	2.656	15.597***	1.531		6.683**	13.953***	2.437	4.976	9.252***	-2.955	7.752***
	(2.688)	(3.080)	(2.979)	(2.444)	(1.949)		(2.864)	(3.710)	(2.026)	(3.164)	(2.754)	(2.219)	(2.510)
FF6	22.872***	16.423***	19.659***	26.345***	17.982***	23.326***		19.849***	18.027***	17.682***	21.410***	21.020***	24.136***
	(2.856)	(2.956)	(3.061)	(2.630)	(2.330)	(2.609)		(3.683)	(2.294)	(3.147)	(2.711)	(2.825)	(2.657)
GKX	42.620***	29.258***	31.246***	49.178***	49.435***	50.618***	47.005***		51.832***	28.642***	46.215***	41.856***	49.196***
	(2.731)	(2.348)	(2.547)	(2.609)	(2.719)	(2.675)	(2.914)		(2.699)	(2.608)	(2.733)	(2.709)	(2.678)
HXZ	9.151***	18.262***	14.440***	17.895***	1.714	6.812***	1.173	20.361***		15.775***	9.025***	6.611**	8.278***
	(2.791)	(3.085)	(3.121)	(2.482)	(1.418)	(1.998)	(2.483)	(3.692)		(3.233)	(2.654)	(2.704)	(2.303)
KNS	24.303***	9.449***	6.801***	36.894***	32.074***	33.985***	30.538***	6.840**	36.825***		36.918***	25.122***	38.132***
	(2.473)	(1.668)	(1.329)	(2.628)	(2.609)	(2.619)	(2.858)	(2.994)	(2.713)		(2.792)	(2.544)	(2.668)
KPS	17.858***	19.199***	22.188***	19.803***	16.180***	17.548***	13.361***	9.258**	14.912***	20.290***		13.256***	12.666***
	(2.875)	(3.100)	(3.112)	(2.630)	(2.657)	(2.639)	(2.852)	(3.632)	(2.579)	(3.233)		(2.787)	(2.291)
NN	13.975***	14.284***	6.903***	25.471***	16.592***	16.122***	18.618***	6.624*	18.788***	6.800**	18.955***		18.445***
	(2.608)	(2.952)	(2.659)	(2.596)	(2.389)	(2.055)	(2.871)	(3.480)	(2.539)	(2.847)	(2.694)		(2.429)
RF	13.744***	16.522***	16.400***	13.812***	8.097***	8.839***	9.425***	9.291**	5.511**	17.631***	3.048	2.993	
	(2.882)	(3.085)	(3.108)	(2.626)	(2.569)	(2.500)	(2.905)	(3.700)	(2.326)	(3.211)	(2.381)	(2.613)	

Table 6: Cross alphas.

Elasticity with hedging.

	BPZ_F (1)	BPZ_L (2)	BSV (3)	CRW (4)	DGU (5)	FF3 (6)	FF6 (7)	GKX (8)	HXZ (9)	KNS (10)	KPS (11)	NN (12)	RF (13)
	Rows correspond to left-hand-side portfolios, columns correspond to right-hand-side portfolios												
BPZ_F		-7.196***	-1.548***	1.748***	-1.904***	-0.754***	-0.268***	-8.626***	0.942***	-3.675***	-1.331***	-0.269***	0.988***
		(0.253)	(0.081)	(0.083)	(0.059)	(0.036)	(0.030)	(0.274)	(0.040)	(0.116)	(0.225)	(0.086)	(0.040)
BPZ_L	7.529***		6.500***	6.996***	7.324***	6.972***	9.798***	-1.739***	6.761***	7.192***	7.103***	7.034***	7.147***
	(0.274)		(0.262)	(0.263)	(0.268)	(0.268)	(0.308)	(0.134)	(0.260)	(0.308)	(0.265)	(0.256)	(0.266)
BSV	2.004***	-1.528***		2.043***	1.634***	2.185***	2.022***	-3.556***	1.858***	-0.742***	2.120***	1.501***	1.839***
	(0.077)	(0.117)		(0.077)	(0.067)	(0.080)	(0.076)	(0.141)	(0.072)	(0.050)	(0.087)	(0.073)	(0.071)
CRW	1.966***	1.795***	2.035***		2.087***	1.793***	1.933***	1.207***	1.645***	2.009***	2.027***	2.099***	1.971***
	(0.085)	(0.081)	(0.086)		(0.088)	(0.081)	(0.085)	(0.072)	(0.076)	(0.085)	(0.087)	(0.086)	(0.085)
DGU	2.351***	0.658***	1.184***	2.612***		4.151***	4.394***	1.203***	6.012***	0.975***	0.372**	2.014***	2.030***
	(0.089)	(0.025)	(0.056)	(0.101)		(0.119)	(0.113)	(0.050)	(0.142)	(0.051)	(0.148)	(0.123)	(0.082)
FF3	0.906***	1.571***	0.150***	1.444***	-0.307***		0.794***	1.195***	1.517***	0.203***	-0.327***	0.158***	0.736***
	(0.028)	(0.052)	(0.021)	(0.058)	(0.023)		(0.023)	(0.037)	(0.048)	(0.016)	(0.096)	(0.059)	(0.023)
FF6	0.734***	-0.806***	0.440***	0.754***	0.215***	0.672***		-0.509***	1.118***	0.204***	-0.432***	0.495***	0.671***
	(0.021)	(0.038)	(0.016)	(0.022)	(0.008)	(0.020)		(0.024)	(0.034)	(0.014)	(0.091)	(0.022)	(0.020)
GKX	9.659***	8.457***	10.512***	9.255***	9.522***	9.346***	10.691***		9.047***	11.229***	8.788***	9.583***	9.480***
	(0.316)	(0.274)	(0.323)	(0.310)	(0.314)	(0.313)	(0.325)		(0.314)	(0.339)	(0.296)	(0.309)	(0.313)
HXZ	-0.012	0.541***	-0.051***	0.740***	-1.319***	-0.430***	-0.760***	1.149***		0.059***	-1.894***	-0.399***	-0.235***
	(0.009)	(0.017)	(0.009)	(0.037)	(0.047)	(0.021)	(0.032)	(0.035)		(0.006)	(0.166)	(0.035)	(0.024)
KNS	3.510***	-2.228***	2.602***	3.364***	3.064***	3.526***	3.758***	-3.814***	3.222***		3.343***	2.682***	3.262***
	(0.126)	(0.158)	(0.090)	(0.121)	(0.113)	(0.123)	(0.127)	(0.152)	(0.117)		(0.122)	(0.108)	(0.118)
KPS	3.514***	3.251***	4.007***	3.891***	3.415***	3.387***	4.386***	0.161	5.634***	3.652***		3.322***	3.112***
	(0.478)	(0.488)	(0.446)	(0.462)	(0.497)	(0.561)	(0.523)	(0.559)	(0.544)	(0.468)		(0.423)	(0.488)
NN	2.243***	-0.354**	0.931***	2.613***	1.156***	1.741***	1.987***	-3.170***	2.293***	0.458***	0.320***		2.121***
	(0.153)	(0.169)	(0.108)	(0.151)	(0.137)	(0.136)	(0.143)	(0.196)	(0.136)	(0.121)	(0.033)		(0.154)
RF	0.957***	4.405***	1.680***	1.814***	-0.527***	0.322***	0.752***	-4.596***	1.561***	2.502***	-0.207	0.520***	
	(0.034)	(0.205)	(0.064)	(0.079)	(0.069)	(0.051)	(0.037)	(0.289)	(0.043)	(0.099)	(0.371)	(0.057)	

Table 7: Elasticity with hedging.

5.6 Table 8, Figure 4, and Table 9: systematic elasticity and aggregate demand

Read:

- Table 8 reports systematic elasticity estimates from RCA/PCA portfolios and shows small average systematic elasticities.
- Figure 4 estimates aggregate demand elasticities from a demand-system model.
- The unscaled log term creates very high elasticity for the smallest stocks, but scaling produces more stable elasticities.
- Table 9 reports goodness-of-fit measures. The LLDM with $c = 1.25$ has the highest overall median R^2 among the reported LLDM specifications, around 0.467.
- The counterfactual equilibrium uses the LLDM with $c = 1.25$, giving average elasticity around 0.81.

Economic message: Even when quantitative strategies are embedded in an equilibrium demand system, aggregate demand remains inelastic.

Table 8
Systematic elasticity estimates.

PCA	BPZ_F (1)	BPZ_L (2)	BSV (3)	CRW (4)	DGU (5)	FF3 (6)	FF6 (7)	GKX (8)	HXZ (9)	KNS (10)	KPS (11)	NN (12)	RF (13)
	Average systematic elasticity												
1/+	0.199***	-0.991***	-0.005	0.326***	-0.224***	-0.170***	-0.028***	0.326***	-0.034***	-0.254***	-0.056**	-0.248***	0.027***
	(0.022)	(0.062)	(0.014)	(0.010)	(0.017)	(0.009)	(0.007)	(0.019)	(0.004)	(0.021)	(0.028)	(0.026)	(0.010)
2/+	0.414***	-0.806***	0.169***	0.479***	-0.063***	0.042***	0.135***	0.453***	0.129***	-0.088***	-0.005	-0.192***	0.145***
	(0.027)	(0.074)	(0.017)	(0.014)	(0.017)	(0.010)	(0.010)	(0.029)	(0.007)	(0.024)	(0.046)	(0.029)	(0.013)
3/+	0.473***	-0.646***	0.202***	0.590***	0.001	0.025**	0.175***	0.576***	0.158***	-0.014	0.063	-0.181***	0.188***
	(0.030)	(0.076)	(0.019)	(0.017)	(0.017)	(0.010)	(0.011)	(0.034)	(0.008)	(0.026)	(0.061)	(0.029)	(0.016)
1/-	0.608***	-0.637***	0.251***	0.680***	0.055**	0.099***	0.227***	0.618***	0.242***	0.010	0.104	-0.203***	0.265***
	(0.037)	(0.093)	(0.024)	(0.020)	(0.022)	(0.012)	(0.012)	(0.042)	(0.012)	(0.033)	(0.084)	(0.033)	(0.020)
2/-	0.289***	-1.207***	-0.017	0.493***	-0.252***	-0.238***	-0.031***	0.455***	-0.021***	-0.321***	-0.046	-0.323***	0.063***
	(0.030)	(0.081)	(0.020)	(0.014)	(0.021)	(0.012)	(0.009)	(0.027)	(0.005)	(0.028)	(0.046)	(0.033)	(0.014)
3/-	0.227***	-1.230***	-0.003	0.381***	-0.266***	-0.164***	-0.025***	0.354***	-0.025***	-0.317***	-0.080**	-0.309***	0.032***
	(0.027)	(0.078)	(0.017)	(0.011)	(0.020)	(0.010)	(0.008)	(0.022)	(0.004)	(0.026)	(0.035)	(0.033)	(0.012)

Value-weighted and winsorized (5th–95th percentiles) average systematic elasticity for six PCA portfolios: 1/+, 2/+, 3/+, 1/−, 2/−, 3/−. Sample: February 1990 to January 2020. Standard errors (in parentheses) are double-clustered by month and stock.*p<0.1; **p<0.05; ***p<0.01.

Table 8: Systematic elasticity estimates.

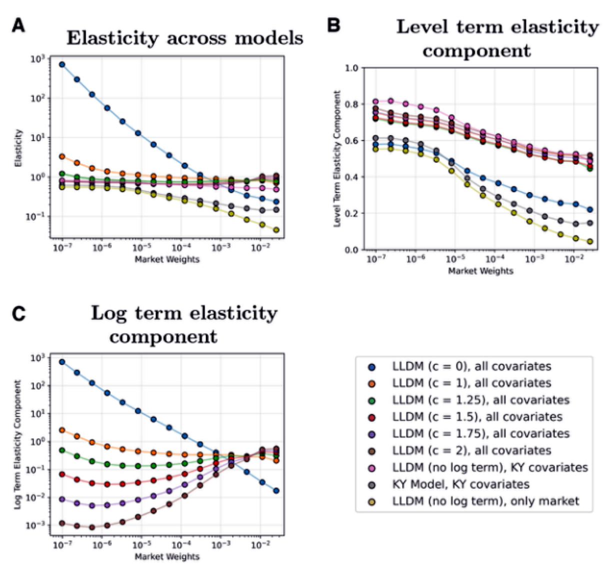

Figure 4
Demand system estimates.
Binned scatterplots. Panel A: Overall elasticity of various models versus market portfolio weights (log-

Figure 4: Demand system estimates.

Table 9
Goodness-of-fit measures.

Model	Covariates	R^2 Quintile 1	2	3	4	5	Overall
KY model	KY	1.000	.997	.976	.888	−.303	.070
LLDM (no log term)	Market	.923	.920	.907	.852	−.385	.029
LLDM (no log term)	KY	.975	.969	.954	.888	−.023	.263
LLDM (c=0)	all	.922	.918	.908	.863	.224	.417
LLDM (c=1)	all	.934	.928	.917	.870	.297	.465
LLDM (c=1.25)	all	.934	.928	.917	.870	.298	.467
LLDM (c=1.5)	all	.932	.927	.916	.870	.295	.466
LLDM (c=1.75)	all	.931	.925	.915	.869	.291	.464
LLDM (c=2)	all	.929	.924	.915	.868	.288	.462

Median R^2 values for models in Figure 4, reported by size quintiles and overall (last column). Quintile R^2 values

Table 9: Goodness-of-fit measures.

5.7 Table A1: model dictionary

Read:

- Table A1 defines the model initialisms used throughout the paper.
- The set includes Fama-French, Hou-Xue-Zhang, Kozak-Nagel-Santosh, Kelly-Pruitt-Su, Gu-Kelly-Xiu, neural networks, and random forests.
- The table also reports how each model forms the function $f(Z_t)$ and the portfolio collapse vector b.

Economic message: The paper's elasticity finding is not about one model. It appears across a broad menu of influential quantitative investment procedures.

Statistical arbitrageur demand models

Initialism	Factor method $f(Z_t)$	MVE collapse method b
BPZ_F	Bryzgalova, Pelger, and Zhu 2025 forest	Bryzgalova, Pelger, and Zhu 2025
BPZ_L	Linear characteristic-weighted portfolios	Bryzgalova, Pelger, and Zhu 2025
BSV	Linear characteristic-weighted portfolios	Brandt, Santa-Clara, and Valkanov 2009
CRW	Linear Chen, Roussanov, and Wang 2022 portfolios	Kozak, Nagel, and Santosh 2020
DGU	Linear characteristic-weighted portfolios	DeMiguel, Garlappi, and Uppal 2009
FF3	Fama and French 1993	Kozak, Nagel, and Santosh 2020
FF6	Fama and French [2015] with Carhart [1997] momentum	Kozak, Nagel, and Santosh 2020
GKX	Gu, Kelly, and Xiu 2021	Bryzgalova, Pelger, and Zhu 2025
HXZ	Hou, Xue, and Zhang 2014	Kozak, Nagel, and Santosh 2020
KNS	Linear characteristic-weighted portfolios	Kozak, Nagel, and Santosh 2020
KPS	Kelly, Pruitt, and Su 2019	Bryzgalova, Pelger, and Zhu 2025
NN	Neural network	$b=1$
RF	Random forest	$b=1$

Initialisms and sources for the 13 statistical arbitrageurs studied. Departures from the original descriptions are noted with justifications. Each model includes (1) a factor model, $f(Z_t)$, and (2) a vector of factor weights, b, collapsing factors into a single MVE portfolio.

Table A1: Statistical arbitrageur demand models.

6 Short summary

- Davis studies the price elasticity of quantitative investment strategies.
- The paper defines statistical arbitrageurs as investors who use asset pricing models to maximize return per unit of volatility.
- The main innovation is to differentiate portfolio weights with respect to prices through price-sensitive predictors.
- A classical mean-variance calibration produces a value-weighted elasticity around 900.
- Thirteen empirical factor and machine-learning models produce much lower elasticities, commonly around 0 to 10.
- The result holds under value-weighting, winsorization, and when short positions are included.
- The key reason is that these models diversify across noisy signals rather than aggressively trading every price change.
- Hedging out other model portfolios leaves pure-alpha portfolios that are still inelastic.
- Systematic elasticity estimates are also low.
- A demand-system equilibrium exercise shows that adding statistical arbitrageurs changes aggregate demand elasticity only minimally.
- The paper therefore challenges the idea that high-Sharpe quantitative strategies automatically provide the elastic demand required by classical no-arbitrage theories.

7 Conclusion

7.1 Core conclusion

The paper concludes that modern cross-sectional quantitative investment strategies are far less price-elastic than classical arbitrage models imply. Even when the models generate substantial out-of-sample alpha and high Sharpe ratios, they do not produce demand curves that move aggressively against price changes.

7.2 Economic intuition

Classical models imagine arbitrageurs buying large amounts when a stock becomes underpriced. Realistic quantitative models do not behave like that. They combine noisy predictors, diversify across many positions, and target Sharpe ratios. Because weights respond only weakly to an individual stock price movement, the resulting demand is inelastic.

7.3 Takeaway

The paper's main lesson is that the existence of sophisticated statistical arbitrageurs does not guarantee elastic market demand. Quantitative alpha may coexist with large price impacts and persistent mispricing because the arbitrageurs themselves are not elastic enough to absorb flows and supply shocks.